

E-Commerce Backend API - Product Requirements Document (PRD)

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Status: Final

Executive Summary

Develop a secure E-Commerce Backend using Python, Django, Django REST Framework, SQLite and JWT Authentication. The project exposes REST APIs for user authentication, category management, product management, cart management and order management.

Business Objectives

Provide secure authentication, efficient product management, shopping cart functionality, order management and scalable REST APIs.

Technologies Used

Python, Django, Django REST Framework (DRF), SQLite, JWT Authentication, Git, GitHub.

Target Users

Customers and Administrators.

Modules

1. User Authentication (JWT)
2. Category Management
3. Product CRUD
4. Cart Management
5. Order Management
6. Admin Panel

Database Tables

Users, Categories, Products, Cart, Cart Items, Orders, Order Items.

REST API Endpoints

POST /api/token/
POST /api/token/refresh/
GET/POST /products/
GET/POST /cart/
GET/POST /orders/

Non Functional Requirements

Secure JWT authentication, API response under 2 seconds, maintainable code structure and scalable backend.

Future Scope

Payment Integration, Inventory Management, Order Tracking, Wishlist, Search & Filter.

Conclusion

This project demonstrates the implementation of a RESTful E-Commerce Backend with authentication, product, cart and order management suitable for academic submission and backend development learning.